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Effect of Passing Plays on Injury Rates in the National Football League

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Abstract

Burke, J, Geller, JS, Perez, JR, Naik, K, Vidal, AF, Baraga, MG, and Kaplan, LD. The effect of passing plays on injury rates in the national football league. *J Strength Cond Res* 35(12S): S1–S4, 2021—The National Football League (NFL) has one of the highest all-cause injury rates in sports, yet our understanding of extrinsic injury risk factors is limited. The objective of this study was to assess the effect of play type on injury incidence in the NFL. We obtained data for every regular season game played during the 2013–2016 seasons from the official NFL game books. There were 2,721 in-game injuries during the 4 seasons examined, with an overall rate of 1.33 injuries per team per game. For statistical analysis, $p < 0.05$ was considered significant. Passing plays conferred significantly higher odds of injury than running plays (odds ratio [OR] 1.4, 95% confidence interval [CI]: 1.3–1.5, $p < 0.0001$). This primarily stems from increased risks in quarterbacks (OR 6.9, 95% CI: 3.6–13.3, $p < 0.0001$), receivers (OR 5.0, 95% CI: 3.7–6.6, $p < 0.0001$), and defensive backs (OR 2.3, 95% CI: 1.9–2.7, $p < 0.0001$). Our study suggests that passing plays confer a greater risk of overall injuries in the NFL when compared with running plays, specifically regarding concussions and core or trunk injuries.

Key Words: injuries, play type, concussion

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Introduction

The National Football League (NFL) has one of the highest all-cause injury rates across major sports, yet our understanding of extrinsic injury risk factors is limited (8). Although there is a large volume of research on specific injuries such as concussions, Anterior Cruciate Ligament injuries, and hamstring strains, we do not have the same understanding of general risk factors contributing to the overall spectrum of NFL injuries (1,2,9,11). One factor associated with increased injury risk is play type (running vs. passing), with previous studies having shown that concussions, hamstring strains, Acromioclavicular joint injuries, and shoulder injuries are more likely to occur during passing plays (2,6,9–11). Conversely, fractures of the distal fibula are more likely to occur on running plays (16).

League-wide, the number of passing plays per team per game has steadily increased over time, which may place players at higher risk for injury. In 1978, only 38% of all plays were pass plays, but this increased to 60% by 2018. In 2018, passer rating (92.9), completion percentage (64.9%), and total touchdown passes (847) all also reached all-time highs (14). With passing plays increasing in frequency, numerous rule changes have also been implemented, adding increased protections for quarterbacks and wide receivers. These include penalties for helmet-to-helmet hits, hits against a defenseless receiver, and landing on the quarterback with the entire weight of a player's body (4). Altogether, added protections for the primary players involved with passing likely contributed to the NFL's passing revolution. As the overall number of passing plays trends upward over time, it is increasingly important to understand the injury implications associated with this trend to minimize player injury risk (14).

The objective of this study was to fill that gap by assessing the effect of play type on incidence of in-game injuries for offensive players in the NFL and to explore how this effect is modified by injury location and player position. We hypothesize that passing plays will result in an increased injury risk for all players, mainly a result of the high speeds that players are moving at when contact occurs. Furthermore, we predict higher injury rates for speed positions, including wide receivers and defensive backs.

Methods

Experimental Approach to the Problem

This study used similar methodology as a previous study undertaken by our group, described in the supplemental file. In brief, we used data extracted from the official NFL game reports for all 32 teams for the 2013–2016 seasons. For the purpose of this study, these reports were used because they detail play type, injuries that occur, weather conditions, position affected, and more. For example, following the description of the results of a play, a typical injury is recorded as follows: "[Player] was injured during the play. His return is [probable]." The University of Miami Institutional Review Board approved this study as exempt, allowing for the use of public data. Informed consent was not required as this was a retrospective study using publicly available data. Player age was not included in the NFL

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gamebooks and thus not recorded in our study. However, it is extremely unlikely that any of the players included were younger than 18 years because the NFL requires players to complete multiple years of college at a minimum before becoming eligible to even be drafted into the NFL. Written permission was obtained from the NFL to use the data derived from gamebooks for epidemiological studies.

Subjects

All in-game injuries reported by the NFL gamebooks were included. Players injured twice in 1 game were recorded as 2 separate injuries. The total number of plays was determined by the recorded number of passing and running plays for each team. Injury location was documented through the following week's injury report found on Foxsports.com. Injury location was categorized into lower extremity, upper extremity, pelvis or hip, trunk or core, head or neck, and concussion. Trunk or core injuries do not include pelvic or groin injuries. Player position was categorized into quarterback, runner (running back and fullback), receiver (wide receiver and tight end), lineman (both defensive and offensive), linebacker, and defensive back. Play type was categorized as run or pass. Special teams and unidentifiable play types (i.e., for difficult to categorize scenarios such as a spiked ball or incomplete plays with a flag) were excluded from analysis. Weather was categorized as dry, wet, or snowy, and temperature was retained as a continuous variable. For indoor games where temperature was not recorded, a value of 70° was assigned, which is a common target temperature for climate-controlled settings. Recorded temperatures ranged from 1 to 94 °F.

Procedures

Statistical analysis was performed by a single biostatistician. Play type and potential confounding variables were characterized using frequency and percentage for categorical variables or mean and SD for continuous variables. Associations with the main outcome were determined in univariable analysis using the chi-squared test or Student's t test as appropriate. For potential confounding variables, associations with play type were also assessed, and variables were included in the final multivariable model if they met a threshold association of $p \leq 0.05$ for both the main outcome and exposure variables. Potential confounding variables were predetermined based on strong associations in the literature.

Statistical Analyses

Logistic regression was used to model the relationship between play type and injuries in multivariable analysis and control for potential confounding variables. Secondary analysis was then conducted in a similar fashion stratifying by position and anatomic location for categories with >100 recorded data points.

Results

Overall, there were 2,721 in-game injuries not occurring during special team plays during the 4 seasons examined, with a median of 1 in-game injuries per team per game (interquartile range 0–2). Passing plays comprised 55.7% of the total plays. On univariable analysis, passing plays were associated with significantly higher odds of injury than running plays (odds ratio [OR] 1.4, 95% confidence interval [CI] 1.2–1.5, $p < 0.0001$) (Table 1).

After controlling for weather conditions, and temperature, passing plays conferred greater odds of injuries than running plays (OR 1.4, 95% CI: 1.3–1.5, $p < 0.0001$) (Table 2). This relationship was not confounded by weather conditions or temperature, with minimal change (<5%) in the primary effect size with removal of each of the variables.

Secondary analyses by injury location demonstrated an even stronger association between pass plays and concussions (OR 1.7, 95% CI: 1.2–2.4, $p = 0.002$) and core or trunk injuries (OR 1.7, 95% CI: 1.1–2.6, $p = 0.01$). There was a smaller but present effect on upper extremity injuries (OR 1.4, 95% CI: 1.1–1.9, $p = 0.02$), and no effect on lower extremity injuries (OR 1.0, 95% CI: 0.9–1.2, $p = 0.70$) (Table 3).

After stratifying by player position, quarterbacks (OR 6.9, 95% CI: 3.6–13.3, $p < 0.0001$), receivers (OR 5.0, 95% CI: 3.7–6.6, $p < 0.0001$), and defensive backs (OR 2.3, 95% CI: 1.9–2.7, $p < 0.0001$) had significantly higher rates of injury with passing plays as compared with running plays, whereas runners (OR 0.4, 95% CI: 0.3–0.6, $p < 0.0001$) and linemen (OR 0.7, 95% CI: 0.6–0.8, $p < 0.0001$) had a decreased risk of injury with passing plays (Table 4).

Table 1

Observed injuries with different game conditions by play type in the regular season, the National Football League, 2013–2016.*

	Running plays	Passing plays	p
Overall	1,023 (6.4)	1,698 (8.4)	<0.0001
Weather condition			<0.0001
Dry	669 (4.2)	1,080 (5.4)	
Wet	53 (0.3)	90 (0.4)	
Snowy	18 (0.1)	16 (0.1)	
Indoors	260 (1.6)	468 (2.3)	
N/A	23 (0.1)	44 (0.2)	
Temperature	67 (54–70)	67 (54–70)	<0.0001

*Continuous variables expressed as median (interquartile range). Categorical variables expressed as frequency (%).

Discussion

This is the first study to demonstrate that, despite rule changes to protect quarterbacks and wide receivers, overall more in-game injuries occur with passing plays than running plays (4). Wide receivers, quarterbacks, and defensive backs were the primary contributors to the increased injury risk, and passing plays seem to confer the greatest risk of concussions and core or trunk injuries. Although previous literature has examined the risk that particular player positions (e.g., wide receivers) have for specific injuries such as concussions, the relationship of passing and

Table 2

Multivariable logistic model of injuries by play type in the regular season, the National Football League, 2013–2016.*

Variable	Odds ratio	95% CI	p
Play type			<0.0001
Passing	1.4	1.3–1.5	
Running (ref)	1		
Weather conditions			<0.0001
Dry	0.8	0.7–0.8	
Wet	0.7	0.6–0.9	
Snow	1.3	0.9–2.0	
N/A	0.3	0.2–0.4	
Indoors (ref)	1		
Temperature	0.9	0.9–1.0	<0.0001

*CI = confidence interval.

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Table 3

Effect in multivariable logistic regression of play type on injury risk by anatomic location in the regular season, the National Football League, 2013–2016.*

Injury location	Injuries during running plays, *n* (%)	Injuries during passing plays, *n* (%)	OR	95% CI	*p*
Concussion	53 (0.3)	111 (0.6)	1.7	1.2–2.4	0.002
Core/trunk	34 (0.2)	71 (0.4)	1.7	1.1–2.6	0.01
Lower extremity	291 (1.8)	375 (1.9)	1.0	0.9–1.2	0.70
Upper extremity	68 (0.4)	121 (0.6)	1.4	1.1–1.9	0.02

*OR = odds ratio; CI = confidence interval.

running plays with total team injuries was to date unexamined. In the setting of steadily increasing passing plays per team per game over the past several decades, our finding support the need to balance player protections with the evolution of the game (14).

After controlling for potential confounding variables including weather conditions and temperature, passing plays conferred a significantly higher risk of in-game injuries than running plays (OR 1.4, 95% CI: 1.3–1.5). By injury location, this seems to have been most significantly seen with concussions (OR 1.7, 95% CI: 1.2–2.4) and core or trunk injuries (OR 1.7, 95% CI: 1.1–2.6). An increased risk of concussions with passing plays aligns with the findings of previous studies, which showed concussions to be most common in positions that tend to be more involved in pass plays such as quarterbacks, wide receivers, and players in the secondary (1,12). Likewise, a "West Coast" style, which is considered to be a pass-heavy offense, has also been found to be associated with higher rates of concussions (15). Higher-speed collisions that occur with passing plays given the increased time of play and spread of the players may contribute to the relationship between passing plays and injuries.

After stratifying by position, the results were intuitive as quarterbacks (OR 6.9, 95% CI: 3.6–13.3), receivers (OR 5.0, 95% CI: 3.7–6.6), and defensive backs (OR 2.3, 95% CI: 1.9–2.7) had the most significant increased risk of injury with passing plays, whereas running backs (OR 0.4, 95% CI: 0.3–0.6) and linemen (OR 0.7, 95% CI: 0.6–0.8) had a higher risk with running plays. This is consistent with previous literature showing higher injury rates with passing plays in wide receivers and cornerbacks, but not running backs (7). Previous work has shown that quarterbacks and wide receivers have greater rates of concussions with passing plays, although in our study the sample size of concussions with these positions was insufficient to conduct an unbiased analysis (15). Both these findings can also be explained by the aforementioned proposed mechanism of high-speed collisions because quarterbacks and wide receivers are among the most vulnerable players to these exposures.

Numerous NFL reforms implemented between 1978 and 2013 have attempted to promote player safety, but policy changes can have unintended consequences. For example, the "Crown-of-the-Helmet Rule," which penalizes players who initiate contact using the top of the helmet, was implemented in 2013 intended to reduce concussion rates. Although this was effective in its goal with defensive players having a 32% decreased rate of concussions after implementation of the policy, an unintended consequence was that in offensive players there was an increase in lower extremity injuries by 34% (3). Similarly, it is possible that the increased player protections for quarterbacks and wide receivers have contributed to the longer-term trend of increased passing plays (14). Given our study's findings, the authors of future studies should examine the direct effects of rule changes on in-game injury risks.

This is the first study to the best of our knowledge to examine overall in-game injury rates by play type because past studies have limited analyses to specific injuries or player types (2,6,9,10). Another key strength of this study is its methodological approach to data extraction. This study collected objective data gathered from official gamebooks to assess the type of play in which each injury in-game occurred. The purpose of gamebooks is not for injury research but rather to assist NFL media personnel in their coverage of games. Therefore, smaller injuries which do not require in-game attention may not be reported to media personnel and therefore would not be recorded in the gamebooks.

Gamebooks are a live, running transcript of NFL games. In addition to recording key details of each play (time of play, key players involved in play, type of play, etc), the gamebooks detail time-outs, penalties, and injuries. Team medical personnel (physician, athletic trainer, etc) report injuries to the media staff, along with the player's projected status for the remainder of the game. This is recorded as: "[Player] was injured during the play. His return is [probable]." Because the injury data rely on reporting from individual teams' medical staff to the media, injuries requiring immediate medical attention, such as more severe injuries and those resulting in stoppage of play, are more likely to be recorded. On the other hand, less severe injuries that do not result in stoppage of play or that are not reported to the medical team by the player until a later time may be missed. This can help explain why the rate of 1.33 injuries per team per game found in our study is fewer than the number of injuries per game reported by the NFL for recent years (5). It is important to note that the goal of this study was to investigate the correlation between play type and in-game injuries. Although gamebooks fail to mention the exact

Table 4

Effect in multivariable logistic regression of play type on injury risk by player position in the regular season, the National Football League, 2013–2016.*

Player position	Injuries during running plays, *n* (%)	Injuries during passing plays, *n* (%)	OR	95% CI	*p*
Quarterback	10 (0.1)	84 (0.4)	6.9	3.6–13.3	<0.0001
Receiver	57 (0.4)	348 (1.7)	5	3.7–6.6	<0.0001
Runner	108 (0.7)	59 (0.3)	0.4	0.3–0.6	<0.0001
Lineman	461 (2.9)	428 (2.1)	0.7	0.6–0.8	<0.0001
Linebacker	177 (1.1)	202 (1.0)	0.9	0.8–1.1	0.39
Defensive back	210 (1.3)	577 (2.9)	2.3	1.9–2.7	<0.0001

*OR = odds ratio, CI confidence interval.

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mechanism, type, and severity of injuries sustained in-game, they provide objective data concerning if and when an injury was reported by the medical staff to the media team as a result of a play, which was critical to the methodology of our study. Our study does not account for injuries that occur outside of games, such as during practice. Furthermore, some in-game injuries may not be reported by players until after the game because athletes often continue to play after suffering minor injuries. This is not the first study to use the NFL gamebooks to examine in-game injury rates because a recent study published by Perez at al. used gamebooks to evaluate injury rates for games played on short rest (i.e., Thursday Night Football) (13).

Our study has several limitations. Our injury rate of 1.33 injuries per team per game is lower than that reported by the NFL in recent years, as discussed above. This is an inherent limitation because smaller injuries may not have been logged in the NFL gamebooks, yet these injuries are still important to consider in our evaluation of the effect of play type on injury rates. Furthermore, non–in-game injuries incurred were not included. Although we believe this is a strength of our study as our main goal was to evaluate in-game injuries by play type, this can also be viewed as a limitation because non–in-game injuries can still influence in-game injuries. Gamebooks do not record the severity of injury, so we are unable to determine whether or not passing plays confer greater risk of more severe injuries than running plays. While it may be somewhat counterintuitive, it is possible that the aforementioned rule changes have decreased the incidence of severe injuries and total games missed despite the overall increase in in-game injuries recorded. We would argue, however, that the increased risk of concussions on passing plays suggests that a higher number of passing plays per game does in fact increase the risk of head injury. Finally, it is possible that increased injury rates could be affected by changes in the average speed and size of players over time.

Future research should further delineate the relationship between play type and injuries, particularly as it relates temporally to the previously discussed rule changes. Furthermore, studies including more granular details about the mechanism and severity of injuries would be useful in determining the need and utility for current and additional player protections. Finally, work to more completely understand risk factors for concussion and methods to reduce their incidence should be continued.

Practical Applications

Passing plays seem to lead to greater numbers of in-game injuries in offensive players in the NFL when compared with running plays. Specifically, passing plays incurred a significantly greater risk of concussions and core or trunk injuries than running plays during the years of 2013–2016. Given the increasing proportion of passing plays in recent years, our results suggest that this trend may be accompanied by a greater risk of injury. Coaches and athletes should consider using fewer passing plays to decrease injury risk. At a minimum, this knowledge should be taken into consideration when planning team practices because our study can help coaches tailor their practices in a way that decreases injury risk. In addition, understanding the epidemiology of NFL injuries incurred on passing plays will help medical professionals tailor their treatments to help athletes to maximize playing time and performance level while minimizing injury risk. Finally, athletic trainers and physical therapists can use this information to formulate workout regimens for players at high-risk positions (i.e., wide receivers) to help them strengthen their core or trunk and potentially prevent or alleviate this type of injury.

REFERENCES

1. Dai JB, Li AY, Haider SF, et al. Effects of game characteristics and player positions on concussion incidence and severity in professional football. *Orthop J Sports Med* 6: 2325967118815448, 2018. doi: 10.1177/2325967118815448.
2. Elliott MC, Zarins B, Powell JW, Kenyon CD. Hamstring muscle strains in professional football players: A 10-year review. *Am J Sports Med* 39: 843–850, 2011.
3. Hanson A, Jolly NA, Peterson J. Safety regulation in professional football: Empirical evidence of intended and unintended consequences. *J Health Econ* 53: 87–99, 2017.
4. Health & Safety Rules Changes. NFL Football Operations. Available at: https://operations.nfl.com/football-ops/nfl-ops-honoring-the-game/health-safety-rules-changes/. Accessed: February 24, 2021.
5. Health & Safety Rules Changes. NFL Football Operations. Available at: https://www.playsmartplaysafe.com/newsroom/reports/2017-injury-data/. Accessed: August 18, 2019.
6. Kelly BT, Barnes RP, Powell JW, Warren RF. Shoulder injuries to quarterbacks in the national football league. *Am J Sports Med* 32: 328–331, 2004.
7. Kraeutler MJ, Belk JW, McCarty EC. The effect of the number of carries on injury risk and subsequent season's performance among running backs in the national football league. *Orthop J Sports Med* 5: 2325967117691941, 2017. doi: 10.1177/2325967117691941.
8. Lawrence DW, Hutchison MG, Comper P. Descriptive epidemiology of musculoskeletal injuries and concussions in the national football league, 2012-2014. *Orthop J Sports Med* 3: 2325967115583653, 2015. doi: 10.1177/2325967115583653.
9. Lessley DJ, Kent RW, Funk JR, et al. Video analysis of reported concussion events in the national football league during the 2015-2016 and 2016-2017 seasons. *Am J Sports Med* 46: 3502–3510, 2018.
10. Lynch TS, Saltzman MD, Ghodasra JH, et al. Acromioclavicular joint injuries in the national football league: Epidemiology and management. *Am J Sports Med* 41: 2904–2908, 2013.
11. Pellman EJ, Powell JW, Viano DC, et al. Concussion in professional football: Epidemiological features of game injuries and review of the literature—Part 3. *Neurosurgery* 54: 81–94, 2004; discussion 94–96.
12. Pellman EJ, Viano DC, Casson IR, Arfken C, Powell J. Concussion in professional football: Injuries involving 7 or more days out—Part 5. *Neurosurgery* 55: 1100–1119, 2004.
13. Perez JR, Burke J, Zalikha AK, et al. The effect of Thursday night games on in-game injury rates in the national football league. *Am J Sports Med* 48: 1999–2003, 2020.
14. Shpigel B, Bill P. How Offense Took over the N.F.L. *The New York Times*, The New York Times, 2019. Available at: https://www.nytimes.com/2019/01/19/sports/nfl-offense-records.html. Accessed February 24, 2021.
15. Teramoto M, Petron DJ, Cross CL, Willick SE. Style of play and rate of concussions in the national football league. *Orthop J Sports Med* 3: 2325967115620365, 2015. doi: 10.1177/2325967115620365.
16. Werner BC, Mack C, Franke K, et al. Distal fibula fractures in national football league athletes. *Orthop J Sports Med* 5: 2325967117726515, 2017. doi: 10.1177/2325967117726515.

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